SYNOPSIS

1. **The Tissue System**
2. **Anatomy of Dicotyledonous and Monocotyledonous Plants**

INTRODUCTION

You can very easily see the structural similarities and variations in the external morphology of the larger living organism, both plants and animals. Similarly, if we were to study the internal structure, one also finds several similarities as well as differences. This chapter introduces you to the internal structure and functional organisation of higher plants. **Study of internal structure of plants is called Anatomy. Plants have cells as the basic unit, cells are organised into tissues and in turn the tissues are organised into organs.** Different organs in a plant show differences in their internal structure. Within angiosperms, the monocots and dicots are also seen to be anatomically different. Internal structures also show adaptations to diverse environments.

The Tissue System - Epidermal, Ground, and Vascular

- The group of cells forming the same function is called **Tissue.**
- The group of various tissue associate to form **Tissue system.**

Types of tissue system

- On the basis of their structure and location, tissue systems can be divided into

i. **Epidermal tissue system**
ii. **Ground tissue system (or) Fundamental tissue**
iii. **Vascular (or) Conducting tissue system**

i. Epidermal Tissue System

- Forms the outermost covering of plant body.
- Epidermal Tissue system consists of

(a) Epidermal cells
(b) Stomata
(c) Epidermal appendages

(a) Epidermal cells

- Epidermal cells are made up of elongated, compactly arranged parenchymatous cells with large vacuole.
- The epidermal cells form outermost layer of the plant body called **Epidermis**.
- Epidermis is single layer.
- The outside of the epidermis is often covered with waxy thick layer called **Cuticle.**
- Cuticle is absent in roots.

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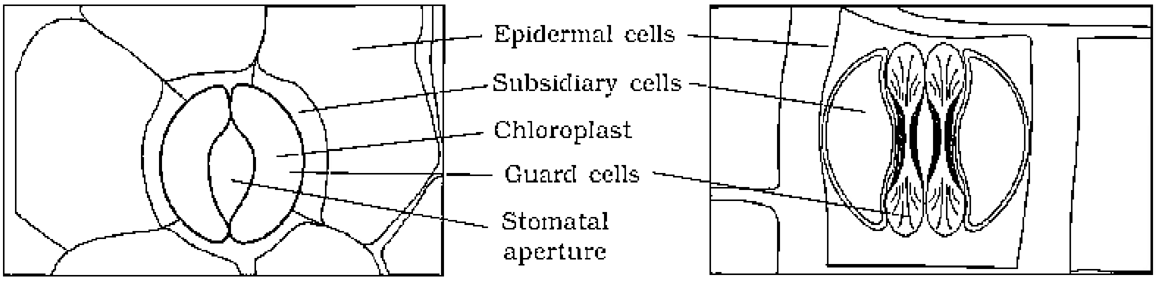
(b) Stomata

- Stomata are structure present in the epidermis of leaves.
- Stomata regulate **transpiration** and **gaseous** exchange.
- Each stoma is composed of two **bean shaped** cells known as **guard cells,** which controls opening and closing of stomata.
- In grasses, the guard cells are **dumb-bell** shaped.
- Guard cells possess chloroplast.
- Guard cells are surrounded by **subsidiary cells** (these cells are actually specialized epidermal cells present in the vicinity of guard cells.)
- The stomatal aperture, guard cells and surrounding subsidiary cells are together called **Stomatal apparatus**.



(a) stomata with bean-shaped guard cells (b) stomata with dumb-bell shaped guard cell

(c) Epidermal appendages (hairs)

- The cells of the epidermis bear a number of hairs like structure.
- If these epidermal hairs are present on stem then they are called **Trichomes** and if they are present on roots then they are called **Root hairs.**
- **Trichomes –** are multicellular, may be branched or unbranched and help in preventing water loss during transpiration.
- **Roots hairs –** are unicellular and help to absorb water and minerals.
- Epidermal appendages may be even secretory.

ii. The Ground Tissue System (or) fundamental tissue

- Except epidermis and vascular bundles, all tissues are part of ground tissue system.
- The ground tissue consists of simple permanent tissues such as parenchyma, sclerenchyma, or collenchyma.
- In primary stem and roots – ground tissue is represented by parenchyma present in cortex, pericycle, pith, and medullary rays
- In leaves - ground tissue is represented by mesophyll (thin-walled chloroplast containing cells)

iii. The Vascular Tissue System (or) conducting tissue system

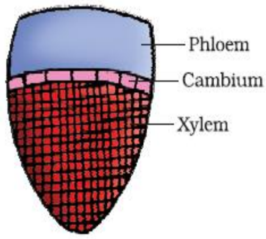
- Vascular tissue system consists of vascular bundles.
- The xylem and phloem together constitute vascular bundle (Vascular bundles = Xylem + Phloem)
- In dicotyledonous plant there is a couple of layer of meristematic cells in between phloem and xylem called **cambium** (or) **vascular cambium** (or) **intrafascicular cambium**

Types of vascular bundle

- There are two types of vascular bundles based on presence or absence of cambium between phloem and xylem.

  (a) Open vascular bundle
  (b) Closed vascular bundle

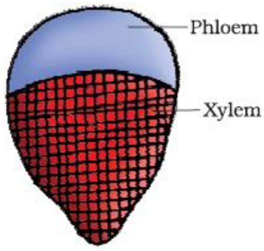
(a) Open vascular bundle



Open vascular bundle

- A vascular bundle is said to be open when cambium is present in between phloem and xylem.
- Cambium is capable of forming secondary growth.
- Open vascular bundle is **present in dicot plants**

(b) Closed vascular bundle



Closed vascular bundle

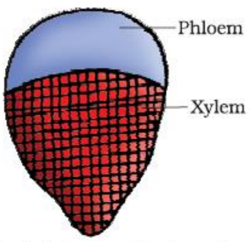
- A vascular bundle is said to be closed when cambium is absent in between xylem and phloem.
- Hence closed vascular bundle is not capable of forming secondary growth.
- Closed vascular bundle is **present in monocot plants**

Arrangement of vascular bundle (xylem and phloem)

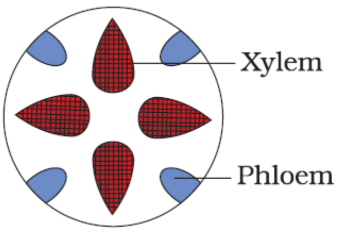
- Xylem and phloem can be arranged in two different kinds of arrangement within a vascular bundle.

a) **Conjoint arrangement of vascular bundle -** when xylem and phloem are present in same radii is called conjoint arrangement of vascular bundle. In Such arrangement of vascular bundle usually has phloem located outer to xylem. Conjoint vascular bundles are **common in stem and leaves.**

b) **Radial arrangement of vascular bundle -** when xylem and phloem are present in alternate manner on different radii is called radial arrangement of vascular bundle. Radial arrangement of vascular bundle are **present in roots**



a. Conjoint vascular bundle



b. Radial vascular bundle

ANATOMY OF DICOTYLEDONOUS AND MONOCOTYLEDONOUS PLANTS

It is convenient to study the transverse sections (T.S) of the mature zones of dicotyledonous and monocotyledonous plants for better understanding of tissue organisation in roots, stems and leaves.

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Transverse Section (T.S) of Dicotyledonous Root (Sunflower root)

- The T.S of dicot root shows following regions namely,

  (i) Epidermis
  (i) Cortex
  (i) Endodermis
  (i) Stele

(i) Epiblema

- Epiblema is the outermost single layer made up of barrel shaped parenchyma cells.
- Many of the epiblemal cells protrude out in the form of unicellular **root hairs.**

(ii) Cortex

- Cortex is the region between epidermis and endodermis
- Cortex consists of many round or oval shaped thin-walled parenchyma cells with intercellular space.

(iii) Endodermis

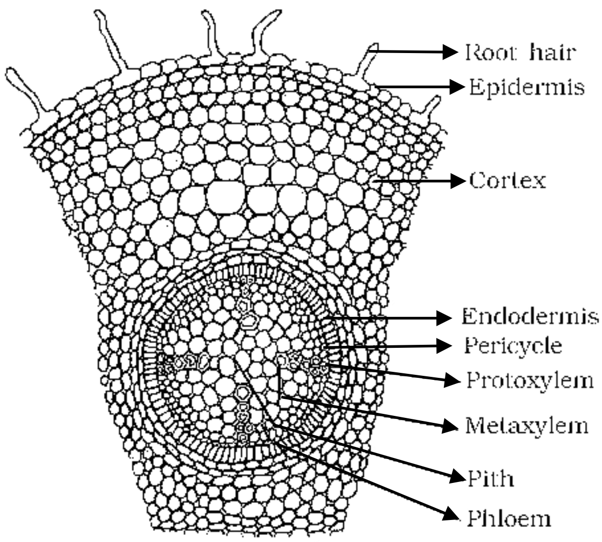
- The innermost layer of the cortex is called **endodermis.**
- Cortex comprises a single layer of barrel-shaped cells without any intercellular spaces.
- The tangential as well as radial walls of the endodermal cells have a deposition of water impermeable, **waxy material-suberin-**in the form of **casparian strips**.

(iv) Stele

- All tissues on the inner side of the endodermis such as **pericycle, vascular bundles** and **pith** constitute the **stele**.

a) Pericycle

- Next to endodermis lie a single layer of thin-walled parenchymatous cells called as **pericycle**.
- Initiation of lateral roots and vascular cambium during the secondary growth takes place in these cells.



T.S of Dicot Root

b) Vascular bundle

- Vascular bundle is **Radial.**
- Xylem is **exarch.**
- There are usually two to four xylem and phloem patches.
- Later, a cambium ring develops between the xylem and phloem.

c) Pith

- The pith is the central axis of root.
- Pith is made up of loosely arranged parenchyma cells with intercellular spaces.
- Pith region is small or inconspicuous

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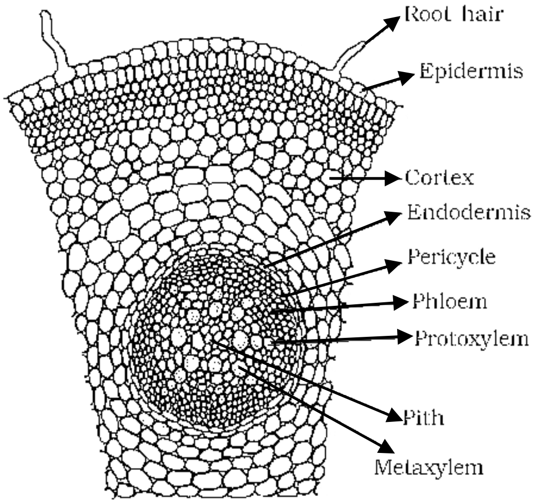
NOTE - The parenchymatous cells which lie between the xylem and the phloem are called **Conjunctive tissue**

Transverse Section (T.S) of Monocot Root

- The T.S of monocot root shows following regions namely,

i) **Epidermis**
ii) **Cortex**
iii) **Endodermis**
iv) **Stele**



T.S of Monocot root

(i) Epidermis

- Epidermis is the outermost single layer made up of barrel shaped parenchyma cells.
- Many of the epidermal cells protrude out in the form of unicellular **root hairs.**

(ii) Cortex

- Cortex is the region between epidermis and endodermis
- Cortex consists of many round or oval shaped thin-walled parenchyma cells with intercellular space.

(iii) Endodermis

- The innermost layer of the cortex is called **endodermis.**
- Cortex comprises a single layer of barrel-shaped cells without any intercellular spaces.
- The tangential as well as radial walls of the endodermal cells have a deposition of water impermeable, **waxy material-suberin-**in the form of **casparian strips**.

(iv) Stele

- All tissues on the inner side of the endodermis such as **pericycle, vascular bundles** and **pith** constitute the **stele**.

a) Pericycle

- Next to endodermis lie a single layer of thin-walled parenchymatous cells called as **pericycle**.

b) Vascular bundle

- Vascular bundle is **Radial.**
- In the monocot root, there are usually more than six **(polyarch)** xylem bundles
- Protoxylem is **exarch.**

c) Pith

- In monocot root pith is large and well developed.
- The pith is the central axis of root.
- Pith is made up of loosely arranged parenchyma cells with intercellular spaces.

NOTE

Monocotyledonous roots do not undergo any secondary growth because of the lack of vascular cambium in between phloem and xylem

Transverse Section of Dicotyledonous Stem

- The T.S of dicot stem shows following regions namely,

(i) **Epidermis**
(ii) **Cortex –** Hypodermis, Cortical layer (ground tissue)
(iii) **Endodermis**
(iv) **Stele**

(i) Epidermis

- Epidermis is the outermost single layer made up of barrel shaped parenchyma cells.
- Epidermis is covered with a thin layer of **cuticle;** it may bear multicellular hairs called **trichomes** and a few **stomata.**

(ii) Cortex

- Cortex cells are arranged in multiple layers between epidermis and endodermis.
- It consists of **two sub-zones.**

a) **Hypodermis: -** Consists of a few layers of collenchymatous cells just below the epidermis, which provide mechanical strength to the young stem.

b) **Cortical layers (ground tissue): -** below hypodermis consist of rounded thin walled parenchymatous cells with conspicuous intercellular spaces.

c) **Endodermis**

- The innermost layer of the cortex is called **endodermis.**
- Endodermis comprises a single layer of barrel-shaped cells without any intercellular spaces.
- The cells of the endodermis are rich in starch grains and the layer is also referred to as the **starch sheath**

(iii) Stele

- All tissues on the inner side of the endodermis such as **pericycle, vascular bundles** and **pith** constitute the **stele**.

a) **Pericycle**

- Pericycle is present on the inner side of the endodermis and above the phloem in the form of **semi-lunar patches of sclerenchyma.**

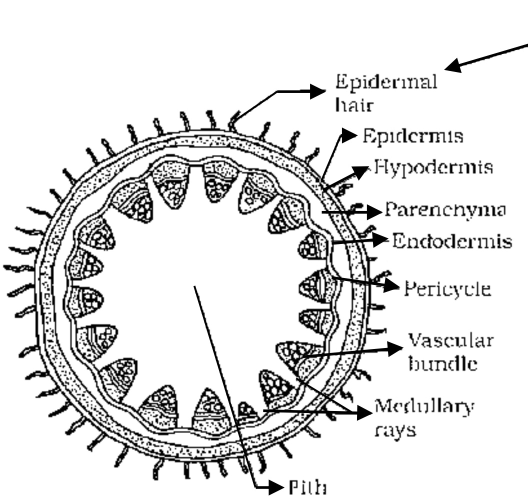
b) **Vascular bundle**

- A large number of **vascular bundles** are arranged in a ring;
- The 'ring' arrangement of vascular bundles is a characteristic of dicot stem.
- Each vascular bundle is **conjoint, open,** and with **endarch protoxylem.**

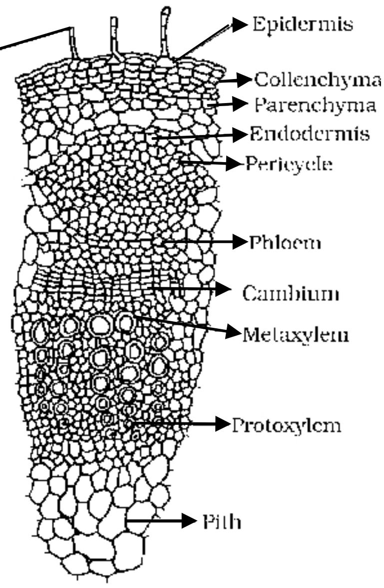
> **NOTE**
> In between the vascular bundles there are few layers of radialy placed parenchymatous cells, which are called as **medullary rays**

c) Pith

- Pith constitutes of a large number of rounded, parenchymatous cells with large intercellular spaces which occupy the central portion of the stem.



T.S of Dicot stem

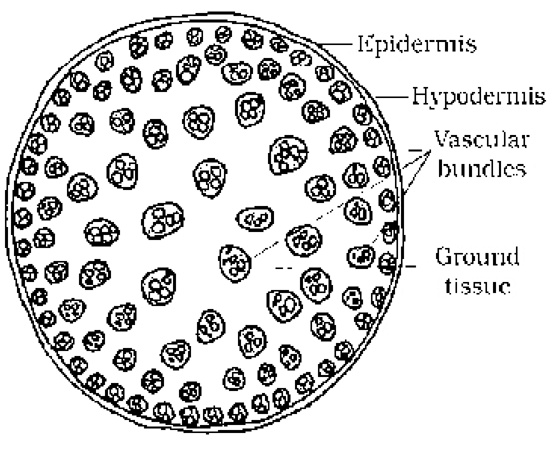


T.S of dicot stem (enlarged portion)

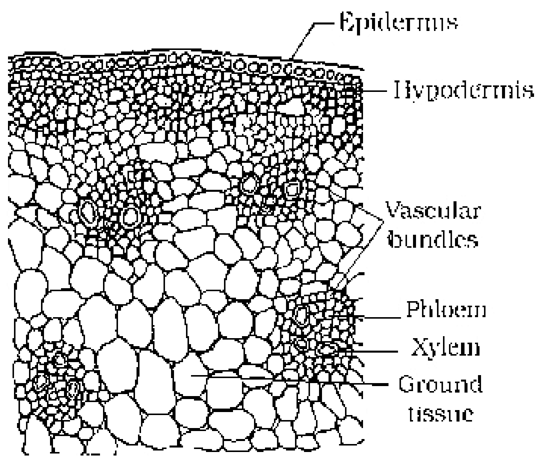
Transverse Section (T.S) of Monocotyledonous Stem

- The T.S of monocot stem shows following regions namely,

  (i) Epidermis
  (ii) Cortex – Hypodermis, Cortical layer (ground tissue)
  (iii) Vascular bundle



T.S of Monocot Stem



T.S of Monocot Stem (enlarged portion)

(i) Epidermis

- Epidermis is the outermost single layer made up of compactly arranged barrel shaped parenchyma cells.

(ii) Cortex

- Cortex cells are arranged in multiple layers below epidermis.
- It consists of **two sub-zones.**

a. **Hypodermis: -** Consists of a few layers of sclerenchymatous cells just below the epidermis, which provide mechanical strength to the young stem.

b. **Cortical layers (ground tissue): -** Is a large, conspicuous region below hypodermis consists of rounded thin walled parenchymatous cells with intercellular space.

(iii) Vascular bundle

- In monocot stem vascular bundles are numerous in number and scattered.
- Each vascular bundle is surrounded by a **sclerenchymatous bundle sheath**.
- In monocot plants vascular bundles are **conjoint** and **closed** with **endarch xylem.**
- Peripheral vascular bundles are generally smaller than the centrally located ones.

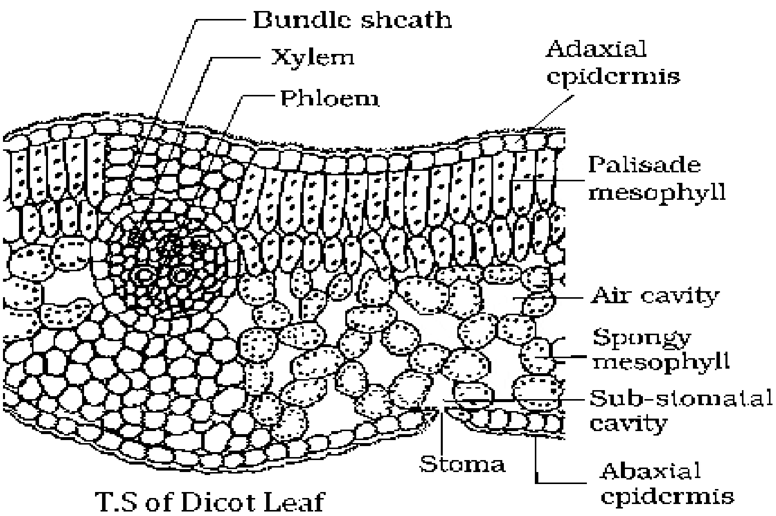
NOTE

- The phloem parenchyma is absent in monocot plants and water containing cavities are present within the vascular bundles.
- Pith is absent in monocot stem

Transverse Section (T.S) of Dicotyledonous (Dorsiventral) Leaf

- The T.S of dicot leaf shows following regions namely,

(i) **Epidermis**
(ii) **Mesophyll**
(iii) **Vascular bundle**



T.S of Dicot Leaf

(i) Epidermis

- The dicot leaf is made up of two epidermis on both upper and lower surface.
- The epidermis which covers upper surface is called as **adaxial epidermis** and lower surface is called as **abaxial epidermis.**
- On both the surface of the leaf has a conspicuous waxy layer called as **cuticle.**
- The abaxial (lower) epidermis generally bears more stomata than the adaxial (upper) epidermis. The latter may even lack stomata.

(ii) Mesophyll

- The tissue between the upper and the lower epidermis is called the **mesophyll**.
- Mesophyll is made up of parenchyma cells which possesses chloroplasts and carry out photosynthesis.
- Mesophyll has two types of cells – the **palisade parenchyma** and the **spongy parenchyma**.
- Palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.
- Spongy parenchyma cells are oval or round shaped and loosely arranged which are situated below the palisade cells and extends to the lower epidermis.
- There are numerous large spaces and air cavities between spongy parenchyma cells.

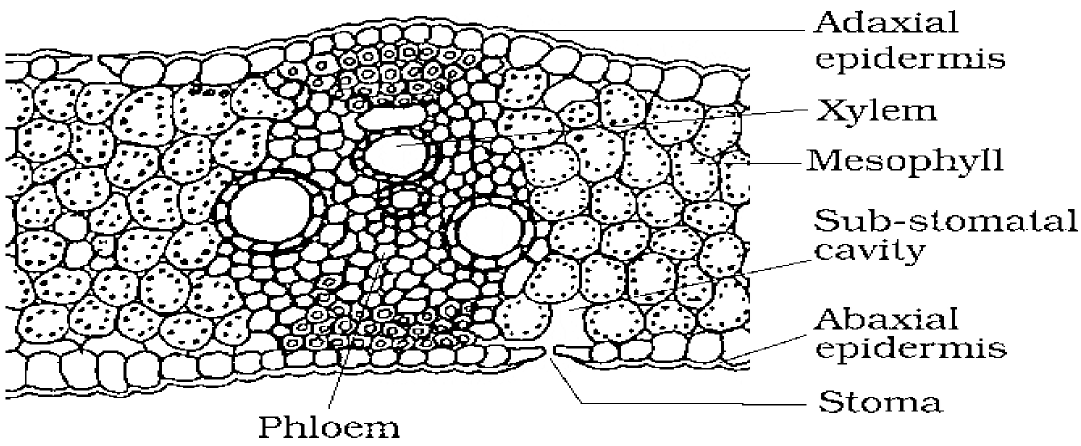
(iii) Vascular bundle

- Vascular bundles can be seen in the veins and the midrib.
- The vascular bundles are surrounded by a layer of parenchymatous cells called as **bundle sheath**.
- Vascular bundle is conjoint and closed

Transverse Section (T.S) of Monocotyledonous (Isobilateral) Leaf

- The T.S of monocot leaf shows following regions namely,

  (i) Epidermis
  (ii) Mesophyll
  (iii) Vascular bundle



T.S of Monocot Leaf

(i) Epidermis

- The monocot leaf is made up of two epidermis on both upper and lower surface.
- The epidermis which covers upper surface is called as **adaxial epidermis** and lower surface is called as **abaxial epidermis.**
- On both the surface of the leaf has a conspicuous waxy layer called as **cuticle.**
- The stomata are present on both the surfaces of the epidermis.

NOTE

In grasses, certain adaxial epidermal cells along with the veins modify themselves into large, empty, colourless cells. These are called **bulliform cells.** The leaf surface is exposed when the bulliform cells absorb water and become **turgid**. When the bulliform cells loose water they become **flaccid** and the leaves curl inwards to minimize water loss.

(ii) Mesophyll

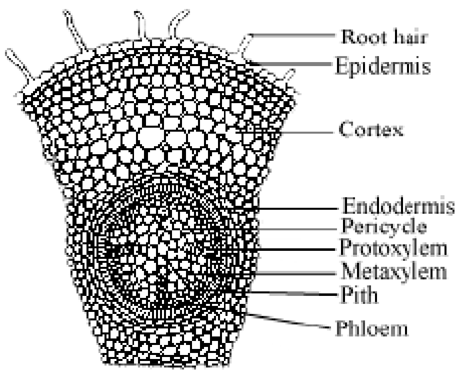
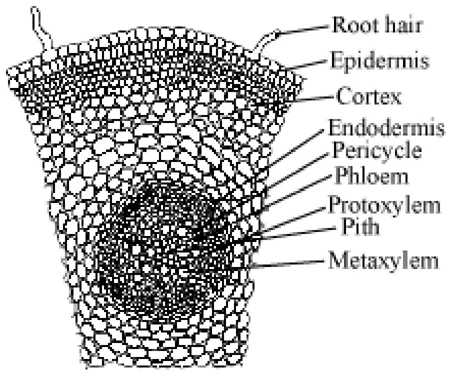
- The tissue between the upper and the lower epidermis is called the **mesophyll**.
- Mesophyll is made up of chloroplast containing parenchyma cells which are not differentiated into palisade and spongy parenchyma.

(iii) Vascular bundle

- The vascular bundles are surrounded by a layer of parenchymatous cells called as **bundle sheath cells**.
- Vascular bundle is conjoint and xylem is endarch

Difference between of Dicot and Monocot Plants

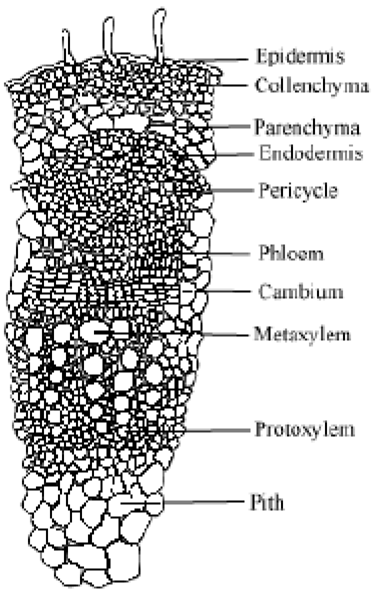
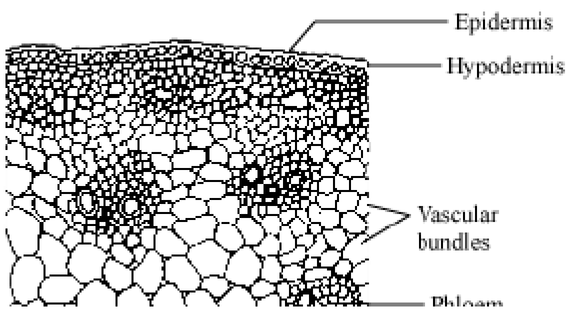
1) Difference between Dicot Root and Monocot Root

Tissue organization	Dicot Root	Monocot Root
Figure		
Epidermis	Has a single layer of epidermal cells, some of which protrude to form root hairs	Has a single layer of epidermal cells, some of which protrude to form root hairs

Cortex	Has several layers of thin-walled parenchymatous cells, with intercellular spaces	Has several layers of thin-walled parenchymatous cells, with intercellular spaces
Endodermis	Single layer of barrel-shaped cells, without intercellular space, and contains Casparian strips (water impermeable layer consisting of waxy suberin)	Single layer of barrel-shaped cells, without intercellular space, and contains Casparian strips (water impermeable layer consisting of waxy suberin)
Pericycle	Has thick-walled parenchyma In these cells, initiation of lateral cambium and vascular bundles responsible for secondary growth takes place.	Secondary growth is absent in monocots.
Pith	Small and inconspicuous	Large and well-developed
Vascular bundle	Single (Monoarch)	More than six (Polyarch)

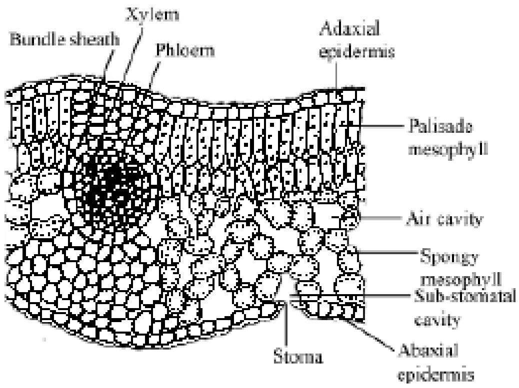
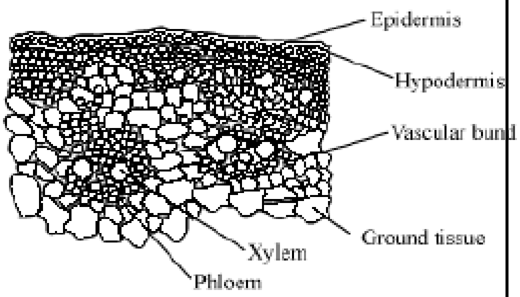
- **Conjunctive tissue –** these are the parenchymatous cells that lie between xylem and phloem. On maturity, cambium rings develop between xylem and phloem.
- **Stele –** structure represented by all tissues on inner side of endodermis such as pericycle, pith, vascular bundle

2) Difference between Dicot Stem and Monocot Stem

Tissue organization	Dicot stem	Monocot stem
Figure		
Epidermis	Covered by cuticle	Covered by cuticle

Cortex (divided into three parts) 1) Hypodermis 2) Cortical layer 3) Endodermis	Hypodermis contains few layers of collenchymatous cells Cortical layer contains parenchymatous cells with conspicuous intercellular spaces Rich in starch	Contains few layers of sclerenchymatous cells - Rich in starch
Pericycle	On the inside of endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma	On the inside of endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma
Vascular bundle	These are arranged in a ring. This arrangement is a characteristic of dicot stem. They are conjoint, open, and have endarch protoxylem.	Vascular bundles are scattered and closed with peripheral bundles being smaller than central. Phloem parenchyma is absent and water containing cavities are present.
Pith	Present	Absent

3) Difference between Dicot leaf and Monocot Leaf

Tissue organization	Dicot leaf (Dorsiventral)	Monocot leaf (isobilateral)
Figure		
Epidermis	Covers both upper (adaxial - bearing less/no stomata) and lower (abaxial - bearing more stomata) surface of leaf and bears a cuticle	Stomata present on both sides
Mesophyll	Differentiated into palisade (has parallel arranged elongated cells) and parenchyma (with loosely arranged oval/round cells); it possesses chlorophyll	Not differentiated into palisade and parenchyma

Vascular Bundle	Present in midrib and veins (reticulate venation); surrounded by bundle sheath cells	Leaves have parallel venation. Bulliform cells (modified epidermal cells) are present along the vein which absorb water and make the cells turgid.